

Munich, 9. Januar 2026

## Profile of Mr. Markus Hopf Dipl.-Ing. (FH)

**Picture:**



**Overview:**

- Dipl.-Ing. Computer Engineering (FH)
- Certified Professional Scrum Master PSM
- Certified Spring Professional
- Sun Certified Java Programmer
- Sun Enterprise Trainer
- Sun Certified Enterprise Architect
- OMG Certified UML Professional
- IBM Certified XML Professional

**Skills:**

- Senior Full Stack Developer
- Cloud Microservice architecture
- Single Page Applications (SPA)
- Jakarta EE, Java, JavaScript, Typescript, Groovy, Kotlin, Scala, Go, Python
- Angular, ReactJs, Wicket
- Spring Boot, Hibernate
- Kafka, NodeJs, npm
- Kubernetes, Docker
- REST API Webservices
- Coordination of Projects

**Availability:**

From **1.4.2026** 100%

**Location:**

**Munich** or Remote.

**Conditions:**

**x,- €** per hour.

**Experience:**

about 30 years

**Birthday:**

13.02.68

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Mobil: +49 172 / 95 94 590  
E-Mail: [info@hopf-it.de](mailto:info@hopf-it.de)  
Homepage: [www.hopf-it.de](http://www.hopf-it.de)

### **Past projects:**

#### **Crypto Key Distribution**

Role: Senior Full Stack Developer  
Branche: Software  
References: Co. Akkodis in Stuttgart/Remote  
Contact person Mr. Edgar Klein Co. Akkodis  
<https://www.akkodis.com/de>

|      |         |   |         |      |           |
|------|---------|---|---------|------|-----------|
| From | 02/2025 | - | 02/2026 | with | 13 months |
|------|---------|---|---------|------|-----------|

REST server JSON encoding based on Java with Spring Boot  
Typescript with Angular Web UI  
Docker container image as build artifact  
Nexus dependencies and target for build artifacts  
Creating and optimizing build and CI/CD pipelines  
Jenkins Multi-Branch Pipeline  
Unit and integration testing with JUnit  
Performance testing and monitoring  
Ensuring code quality through SonarQube analysis  
Optimizing software architecture and deployment processes  
PKI Key Generation  
Java 21 Spring Boot Applications  
JHipster with JDL Schema Files  
Ansible for Process Automation  
SOAP Interface for data exchange  
Message signature with Openssl  
Zertificates and PKCS12-Files  
Liquibase Database Migrations  
Podman Container and Image Management  
LDAP Access over Keycloak  
Postgres as relational Datenbank  
Sonar Quality Checks  
E2E Integration Tests  
Key Management for Satellite Communication  
Role-based Access Control - RBAC  
Provider for critical Infrastructures  
Management of Security Modules  
Multi-Factor Authentication (MFA)  
Crypto know-how  
Key distribution and PKI

Satellite navigation  
 Requirements analysis  
 Agile development of software projects  
 Creation of concepts and requirements  
 Functional software testing  
 Reducing test and development cycles  
 Requirements management, capturing requirements

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Unix, Linux, Windows                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Datenbases                      | PostgreSQL, MariaDB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Developmenttools                | IntelliJ, Visual Studio Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Tools                           | Angular, @angular/material, bootstrap, eslint, TypeScript, webpack, router, SOAP, JHipster, Ansible, Python, pip, SSH Keys, Openssl, PKCS12, keytool, junit, Java, JakartaEE, JavaScript, podman, Docker, Keycloak, FreeIPA, OpenLDAP, Kerberos, Postgres, Spring Boot, Znuyn, IAM, MFA, RedHat, RHEL, Linux, Windows, JIRA, Bitbucket, Exceed TurboX, Jenkins, Sonar, Sonarqube, jacoco, Confluence, checkstyle, archunit, prometheus, JWT, Websockets, Grafana, Logstash, Graylog, checkmk, SNMP, Nexus, gradle, groovy, kotlin, fasterxml, github, vaadin, apache commons, commons, netty, swagger, undertow, jakarta, joda-time, bouncycastle, IntelliJ, jetbrains, eclipse, hibernate, hamcrest, mapstruct, mockito, seleniumhq, slf4j, springframework, wsdl, nodejs, node, npm, Liquibase, REST, JSON, JavaScript, Git, CI/CD, Spring Actuator, Java 21 Streams API, Spring Data, freeIPA, LDAP |

## Bank Migration - Atruvia

Role: Technical Project Management

Branche: Bank

References: Co. Zühlke AG in London/Remote

Contact person Mr. Kris Hubert, Mr. Tommaso Solcia

<https://www.amass-technology.com>

<https://www.zuehlke.com>

|      |         |   |         |      |          |
|------|---------|---|---------|------|----------|
| From | 10/2024 | - | 02/2025 | with | 5 months |
|------|---------|---|---------|------|----------|

Clarification of minimum requirements  
 Relocation of servers to a data center  
 Data center commissioning  
 Project Management Coordination Tasks  
 Migrationproject

- IT-Architektur
- Meeting Planung
- Project Timeplan
- Covering all Customer Requests
- Review Documentation
- Evaluation of Test Results
- Suggestions for Improvements and Enhancements
- Internal Coordination
- Providing continuous Updates
- Report Projectstatus
- Data Transfer and Integration
- Credit Bank Migration for Peridot
- Peridot Customer Migration to Atrivua

|                                 |                                       |
|---------------------------------|---------------------------------------|
| Realized with: Operatingsystems | Windows, Linux                        |
| Datenbases                      | H2, Oracle                            |
| Developmenttools                | Eclipse, IntelliJ, Visual Studio Code |
| Tools                           | maven, IBM                            |

## Mission Control Panel - FELIS

Role: Senior Full Stack Developer  
 Branche: Public Sector, Police  
 References: Co. T-Systems in Berlin/Remote  
 Contact person Mr. Tim Zett, Mr. Benjamin Zeiss, Mr. Frederic Mehl  
<http://www.t-systems-ifs.com>

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|------|---------|---|---------|------|-----------|
| from | 04/2023 | - | 09/2024 | with | 18 months |
|------|---------|---|---------|------|-----------|

- Asynchronous Microservices Landscape
- 30 Microservices for Front- and Backend
- Reactive, resilient and scalable Applications
- Consequent Usage of Reactive Programming Paradigms
- RxJava Reactive Extensions for the JVM
- Redux Frontend Component Store with NgRx
- NgRx Components
- Callbacks, Promises, Futures and Kotlin Coroutines
- Event Bus with Non-Blocking Verticals
- Asynchronous Event Loop
- WebSockets Server-sent Events
- Usage of Fluent API
- Event Sourcing Pattern
- CQRS and Saga Pattern

Eclipse Vert.x Toolkit  
Kendo and Angular Material Design UI Components  
RESTful URL Design  
Vert.x-Web for RESTful HTTP Microservices  
AMQP Messaging with Apache Kafka  
Migration of Vert.x to Kafka  
Distributed Clustering  
Asynchronous Data and Event Streams  
Implementing Back-Pressure Streams  
Virtualised Environments and Containers  
Docker and podman Containerization  
One Time Password - Multi-Factor Authentication  
JWT/OAuth2 based Authorization  
CORS - Cross Origin Resource Sharing  
CSRF - Cross Site Request Forgery  
Zookeeper Cluster Manager  
Monorepo Usage with Nx  
Keycloak IAM Roles and Rights  
Cloud Virtualization-Plattform VMware vSphere  
DevOps Gitlab Pipeline  
API-Gateway  
Python and Robot for Data migration  
AWS S3 File Storage  
Development of a 110 Control-Panel for the Police  
Disposition of Missions  
Policeman Management  
Tracing Management  
Mission Protocol and Management  
Location-based Map with Vectordata  
Creation of Requirements and Concepts  
Creation of Administration Guides  
Team Lead

|                                 |                                         |
|---------------------------------|-----------------------------------------|
| Realized with: Operatingsystems | AWS, Windows, Linux                     |
| Datenbases                      | H2, MariaDB, MsSql, MongoDB, PostgreSQL |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developmenttools<br>Tools | Eclipse, IntelliJ, Visual Studio Code,<br>Jakarta EE, Powershell, maven, gradle, ant,<br>Infinispan, Hazelcast, Vert.x4, Verticle, SLFj4,<br>jackson, netty, logback, assertj, TCP/IP, Http,<br>JUnit5, VisualVM, jMeter, RxJava, RxJS,<br>ReactiveX, JavaScript, REST Assured, Docker<br>Compose, Minikube, Apache Kafka, Apache<br>FreeMarker, Thymeleaf, Pebble, Rocker, X-XSRF-<br>TOKEN, AJAX, XSS, Consul, Zookeeper,<br>Kubernetes, Skaffold, podman, Redis, GraphQL,<br>Failsafe, surefire-plugin, Circuit Breaker, Node.js,<br>SockJS, Prometheus, Micrometer, Grafana,<br>ZooKeeper, Ansible, helm, Terraform,<br>GraphHopper, Gitlab, Node18, Java20, Kendo,<br>Angular Material Design, Angular16, Angular Nx,<br>Jest, Karma, Reducer, Robot Framework, Taurus,<br>Hibernate, DozerMapper, Sonar,<br>SOAP, Liquibase, VMware vSphere, Keycloak,<br>IAM, JWT, OpenID Connect /OAuth2,<br>Websockets, nrwl/nx, nx.dev, Cypress, TypeScript,<br>JavaScript, Html, Css, Atlassian, Jira, Confluence,<br>Bamboo, draw.io, Fileon and ColorZilla |
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## Online Banking - Customer Focus

Role: Senior Full Stack Developer

Branche: Bank

References: Co. Atruvia/attempo in München

<http://www.atruvia.com>

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|------|---------|-----------|------|-----------|
| from | 02/2022 | - 03/2023 | with | 14 months |
|------|---------|-----------|------|-----------|

Service Cut Design

Conception, Consulting and Maintain

Implementation in all Layers

Architecture Diagrams with ADOIT and ArchiMate

Java Spring Boot Microservices

Angular Single Page Application (SPA)

10 Microservices for Front- and Backend

TypeScript, Java, JavaScript, Groovy

RESTful Microservice Architecture

REST API Versioning

Red Hat Open Shift Cloud Environment

Docker Container into Kubernetes Cloud Environment

Jenkins CI/CD-Pipelines

Infrastruktur as Code (IaC)

JSON Web Token (JWT), Bearer- Authorization-Header

Usage of the Omnikanalplattform OKP  
 Implementation of Java Banking Framework JBF  
 Kubernetes Commands and Configuration  
 Quality Tasks with Sonar and Linting  
 Multi Maven Projects  
 Nexus Artifact Repository  
 JSON Data Communication  
 Alerting, Health Checks  
 Postman, Swagger, OpenAPI  
 Redis In Memory Cache  
 Host Connection and COBOL Copybook Integration  
 Testautomation, Performance Analysis, Resilience  
 Cypress and Karma Tests  
 Sleuth, Zipkin, Opentracing, Dynatrace  
 Grafana and Dynatrace Monitoring  
 Spring Boot, Spring Data, Spring Cloud OpenFeign  
 Release and Feature Toggles  
 Spring Expression Language  
 NodeJs JavaScript Backend Development  
 JavaScript NodeJs Tests with Mocha and Chai  
 NGINX Configuration  
 Soap Communication Protocol  
 Kubernetes Secrets Configuration  
 Semantic Versioning  
 DevOps with CI/CD Jenkins Pipelines  
 Unix Shell Scripts with bash  
 Web Performance Analysis Tools  
 Data Exchange with WSDL and SOAP  
 JMeter Heavyweight Tests  
 Development Parts for Online Banking Platform  
 Browser and Mobil based Banking as Single Page Application (SPA)  
 Creation of Concepts  
 Creation of Administration Guides  
 Reviews of Sourcecode  
 Support of Developers  
 Coordination Dev and DevOps Tasks  
 Remote Communication over MS-Teams  
 Agile Development in Scrum Team and SAFe  
 Grooming, Planning, Retrospective, GIP  
 Communication in English and German  
 OnlineBanking and VR Banking App

|                                               |                                                  |
|-----------------------------------------------|--------------------------------------------------|
| Realized with: Operatingsystems<br>Datenbases | Red Hat Open Shift, Windows, Linux<br>H2, Oracle |
|-----------------------------------------------|--------------------------------------------------|

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|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developmenttools<br>Tools | Eclipse, STS, IntelliJ, Visual Studio Code<br>Powershell, ADOIT, ArchiMate, LeanIX, Mapstruct,<br>Hazlecast, Java 17, JUnit 5, Jupiter, Lighthouse,<br>web.dev, requestmap.webperf.tools, webpack-<br>bundle-analyzer, WebP, Angular 15, Postman,<br>newman, Swagger, Git, Bitbucket, Jenkins, OTC,<br>Red Hat OpenShift, AWS, Kubernetes, NGINX,<br>Ingress, JMeter, Kafka, Teams, TypeScript,<br>Groovy, JavaScript, Jasmine, Cypress, Mocha,<br>Chai, Karma, Gherkin, cucumber, mural,<br>xrepository, Sleuth, Zipkin, Opentracing,<br>Dynatrace, Spring Boot, Spring Data, Docker,<br>JPA, Hibernate Envers, LogBack, Log4j2, JWT,<br>Maven, Nexus, Sonar, Lint, Hystrix, H2, Oracle,<br>MsSQL, PostGre, Liquibase, NodeJs, npm, yarn,<br>Checkstyle, curl, JIRA, OWASP, Kubectrl, yaml,<br>Notepad++, GraphQL, SASS/SCSS, SmartBear<br>SoapUi, Citrix, SourceTree, SmartGit, Lombok,<br>@angular/material, SQLDeveloper, Nx, ngx/store,<br>@ngrx/component-store,, IBM<br>RecordClassGenerator, Redis, Grafana,<br>Prometheus, Istio, kibana, Confluence, JIRA,<br>OpenAPI, ReadyAPI, planitpoker, electron,<br>conceptboard.com, miro, wiremock, lucidchart,<br>arc42, Sketch MeaXure |
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## Shared projects for judiciary - GeFa

Role: Technical Architect

Branche: Public Sector

References: Co. .msg in München/Ismaning

Contact person Mr. Shinja Strasser, Mr. Helmut Lämmermeier

<http://www.msg.de>

<http://www.gefa-justiz.de>

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|------|---------|-----------|------|----------|
| from | 08/2021 | - 02/2022 | with | 7 months |
|------|---------|-----------|------|----------|

Architecture Responsibility and Service Cut  
Domain based Microservices  
Conception of Cross cutting themes  
BSI Security-Requirements  
Analysis of None-Functional-Requirements  
Conception, Maintain and Consulting  
Creation of Architecture Concepts  
Greenfield Project  
Communication with Senior Full stack Devs  
Angular Single Page Application (SPA)



Browser based Court-WebApp  
Approx. 30 Microservices for Front- and Backend  
TypeScript, Java, JavaScript, Groovy  
RESTful Microservice Architecture  
REST API Versioning  
Framework for Standardization of Microservices  
Open Telekom Cloud Environment OTC  
Docker Container into Kubernetes Cloud Environment  
Jenkins CI/CD-Pipelines  
JSON Web Token (JWT), Bearer- Authorization-Header  
Kubernetes Commands and Configuration  
Relational SQL Scripts  
Database Connections and Liquibase Migrations Scripts  
Hystrix latency for Fault Tolerance  
CircuitBreaker, FeignClients, SoapClients  
Quality Tasks with Sonar and Linting  
Multi Maven Projects  
Nexus Artifact Repository  
JSON Data Communication  
Alerting, Health Checks  
Postman, Swagger, OpenAPI  
Testautomation, Performanceanalysis, Resilience  
Cypress and Karma Tests  
Sleuth, Zipkin, Opentracing, Dynatrace  
Monitoring, Security  
Spring Boot, Spring Data, Spring Cloud OpenFeign  
Release and Feature Toggles with trunk-based-development  
Spring Expression Language  
NodeJs JavaScript Backend Development  
JavaScript NodeJs Tests with Mocha and Chai  
Ingress Service Discovery  
NGINX Configuration  
Soap Communication Protocol  
Kubernetes Secrets Configuration  
Semantic Versioning  
Angular State Management with ngxs/store, Actions and Reducer  
Angular Material Design UI Components  
DevOps with CI/CD Jenkins Pipelines  
Shell Scripts  
Web Performance Analysis Tools  
Data Exchange with WSDL and SOAP  
JMeter Heavyweight Tests  
Operation of Certification Authority (CA) for PKI  
Usage of Public-Key-Infrastructur (PKI)

Meetings with business domain experts  
 Development of a Court-Software for the German Justice  
 Browser based Court-App as Single Page Application (SPA)  
 Ca. 100 involved Persons in Development  
 Reviews of Sourcecode  
 Coordination of Devs and DevOps Tasks  
 Agile Development in a Scrum Team and SAFe  
 Grooming, Planning, Retrospective  
 Communication in English and German

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Open Telekom Cloud (OTC) , Windows, Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Datenbases                      | H2, Oracle, MsSQL, PostGre                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Developmenttools                | Eclipse, STS, IntelliJ, Visual Studio Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Tools                           | ArchiMate, LeanIX, Mapstruct, Hazlecast, Java 11,<br>Lighthouse, web.dev, requestmap.webperf.tools,<br>webpack-bundle-analyzer, WebP, Angular 13,<br>Postman, newman, Swagger, OpenAPI, Git,<br>Bitbucket, Jenkins, OTC, Kubernetes, NGINX,<br>Ingress, JMeter, Teams, WebEx, Angular,<br>TypeScript, Groovy, Java, JavaScript, Jasmine,<br>Cypress, Mocha, Chai, Karma, Gherkin,<br>cucumber, mural, xrepository, XJustiz, Sleuth,<br>Zipkin, Opentracing, Dynatrace, Spring Boot,<br>Spring Data, Docker, JPA, Hibernate Envers,<br>LogBack, Log4j2, JWT, Maven, Nexus, Sonar,<br>Lint, Hystrix, H2, Oracle, MsSQL, PostGre,<br>Liquibase, NodeJs, npm, yarn, Checkstyle, curl,<br>JIRA, OWASP, Kubectrl, yaml, Notepad++,<br>GraphQL, SASS/SCSS, SmartBear SoapUi, Citrix,<br>SourceTree, SmartGit, Lombok,<br>@angular/material, SQLDeveloper, Nx, ngx/store,<br>Prometheus, Eclipse, STS, IntelliJ, kibana,<br>Confluence, JIRA, coyo |

## Housing-saving Calculator and Taxes

Role: Senior Full Stack Developer  
 Branche: Finance  
 References: Co. Fiducia GAD in München/Aschheim  
 Contact person Mr. Matthias Leukert, Mr Erich Burggraf, Mr. Martin Hahn  
<http://www.fiduciagad.de>

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|------|---------|-----------|------|-----------|
| from | 12/2019 | - 08/2021 | with | 20 months |
|------|---------|-----------|------|-----------|

Angular Single Page Application - SPA  
 12 Microservices for Front- and Backend  
 TypeScript, Java, JavaScript, Groovy  
 RESTful Microservice Architectur

REST API Versioning  
Framework for Standardization of Microservices  
Red Hat OpenShift Cloud Environment  
Red Hat Enterprise Linux (RHEL) Systemadministration  
Docker Container in Azure Cloud Environment  
JSON Web Token (JWT), Bearer- Authorization-Header  
Kubernetes Commands and Configuration  
Relationale SQL Scripts  
Database connections and Flyway Migrations Scripts  
Hystrix latency for Fault Tolerance  
CircuitBreaker, FeignClients, SoapClients  
Quality with Sonar and Linting  
Multi Maven Projects, Gradle Build Scripts  
Nexus Artifact Repository  
AMQP Messaging - Advanced Message Queuing Protocol  
Usage of the Omnikanalplattform OKP  
Implementation of Java Banking Framework JBF  
JSON Data Communication  
Alerting, Health Checks  
Postman, Swagger, OpenAPI  
Camunda Process-Engine  
Camunda BPMN-Workflows and DMN-Decisions  
Testautomation, Performanceanalyze, Resilienz  
Selenium, Fitnesse, Protractor and Karma Tests  
Zipkin, Opentracing, Dynatrace  
Monitoring, Security  
Spring Boot, Spring Data, Spring Cloud OpenFeign  
Release and Feature Toggles with trunk-based-development  
Spring Expression Language  
NodeJs JavaScript Backend Development  
JavaScript NodeJs Tests with Mocha and Chai  
Consol Service Discovery  
Soap Kommunikations Protokoll  
Openshipt Secrets Konfiguration  
Semantic Versioning  
Angular State Management with ngxs/store, Actions and Reducer  
Angular Material Design UI Components  
Migration towards Angular 11  
JQ Json Parser  
DevOps with OpenShift CI/CD Jenkins Pipelines  
Shell Scripts  
Web Perfomance Analyse Tools  
Data exchange with WSDL and SOAP  
JMeter Robustness Tests

Developing of a Housing-saving Calculator  
 Enhancement of the Bank Desktop - BAP  
 All kind of taxes sheets  
 REST Interfaces tou BAP, KAPST and agree21  
 Meetings with buisness domain experts  
 Reviews of Sourcecode  
 Dev and DevOps Tasks  
 Remote Communication over Skype and Gotomeeting  
 Agile Development in a Scrum Team and SAFe  
 Grooming, Planning, Retrospective  
 OnlineBanking and VR Banking App  
 Communication in English and German

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|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Azure Cloud, Windows, Linux, Ubuntu KDE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Datenbases                      | H2, Oracle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Developmenttools                | Mapstruct, Hazlecast, Java 11, miro.com,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Tools                           | HashiCorp Consul, Lighthouse, web.dev,<br>requestmap.webperf.tools, webpack-bundle-<br>analyzer, WebP, Angular 11, Postman, newman,<br>Swagger, OpenAPI, Git, Bitbucket, Jenkins, Red<br>Hat OpenShift, Azure, Camunda, Spin, JMeter,<br>IBM Lotus Notes, Teams, GotoMeeting, Angular,<br>TypeScript, Groovy, Java, JavaScript, Selenium,<br>Jasmine, Protractor, Mocha, Chai, Karma,<br>Fittesse, Zipkin, Opentracing, Dynatrace, Spring<br>Boot, Spring Data, Kubernetes, Docker, AMQP<br>Messaging, JPA, Hibernate, Log4j2, JWT, Maven,<br>Gradle, Nexus, Sonar, Lint, Hystrix, H2, Oracle,<br>SQL, Flyway, NodeJs, npm, yarn, Checkstyle, curl,<br>JIRA, OWASP, Kubectrl, yaml, Notepad++,<br>GraphQL, SASS/SCSS, SmartBear SoapUi, Citrix,<br>Telegram, SourceTree, SmartGit, Lombok,<br>@angular/material, SQLDeveloper, ngxs/store,<br>allure, IKESA, VR-NetKey, aqua, Grafana,<br>Prometheus, funretro.io, IntelliJ, consul.io, gson,<br>kibana |

## Digital Marketplace - G2M

Role: Full Stack Developer

Branche: Telecommunication

References: Co. Rohde & Schwarz in München

Contact person Mr. Patrick Bauriedl, Mr. Ivan Kostov, Mr. Christ. Maurer

<http://www.rohde-schwarz.com>

|      |         |           |      |          |
|------|---------|-----------|------|----------|
| from | 09/2019 | - 12/2019 | with | 4 months |
|------|---------|-----------|------|----------|

Development of Digital Marketplace Applications

WebShop Development with Angular and NodeJs  
Backend and Frontend with TypeScript and JavaScript  
Packet Manager npm and yarn  
Azure Cloud based PaaS  
Docker based MicroServices  
Docker Compose multi-container solutions  
Harbor Registry for Docker images  
Git Hooks with Husky  
Red Hat OpenShift Pods  
OKD Jenkins Pipelines and Clusters  
Administration of OpenShift Clusters  
OC and Kubectl CLI commands  
Yaml Templates as OpenShift Service Definition  
Azure Service Bus for asynchrones Messaging  
REST API Routes with Versioning  
OpenAPI, Swagger Documentation  
Authentification over SAML Service  
NoSQL Datenbase MongoDB  
Configuration as Code with Jenkinsfiles  
CI/CD Pipeline  
Distributed Tracing, OpenTracing  
OWASP and Lizenze checks  
CSS Styles and Html Templates  
Jasmine, Karma and Robot Tests  
Test with Robot Framework and RIDE Suites  
Python Scripte for Test Automation  
Sonar and Checkstyle Coverage  
Dev and DevOps Duties  
Development of a generic Webshop  
Shop Extensibility for cloud products  
Go-2-Market Applications - G2M  
Identity and Access Management - IAM  
Remote Communication over GotoMeeting  
Agile, remote Scrum Team  
Grooming, Planning, Retrospective  
Project language English

|                                 |                                         |
|---------------------------------|-----------------------------------------|
| Realized with: Operatingsystems | Azure Cloud, Windows, Linux, Ubuntu KDE |
| Datenbases                      | MongoDB, CosmosDB, mySQL                |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developmenttools<br>Tools | Red Hat OpenShift, Azure, MongoDB, Docker, Visual Studio Code, Notepad++, Citrix, Chrome DevTools, TypeScript, Angular, Jasmine, Karma, Robot Tests, ESLint, NodeJs, npm, yarn, PMD, Checkstyle, Sonar, JIRA, Docker, Proxyfier, Postman, Hyper-V, VMware, Skype, git, bash, zipkin, jaeger, depcheck, module-structure, retire.js, OWASP, Harbor, CosmosDB, Husky, OKD, OC, Kubernetes, Kubectl, Keycloak, MockServer, Ubuntu KDE, Kate, Dolphin, robotframework, RIDE, Python, pip, MarkdownViewer++, Markdown Extension, json, yaml, JWT, APISimulator, draw.io, Studio 3T, OpenAPI, Swagger |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Siemens Corporate Entitlement Service - CES

Role: Full Stack Developer

Branche: Security

References: Co. Siemens in München

Contact person Mr. Siegfried Eichner, Mr. Binh Le Nguyen, Mr. Erik Bergjohann, Mrs. Elena Thiel

<http://www.siemens.com>

|      |         |   |         |      |           |
|------|---------|---|---------|------|-----------|
| from | 06/2018 | - | 09/2019 | with | 16 months |
|------|---------|---|---------|------|-----------|

- Implementation of Spring Microservices
- Spring Boot and Spring Batch Applications
- Spring Core, Security
- Spring Eureka Multi Instanz Architecture
- Spring Cloud Netflix, Eureka, MVC, Thymeleaf
- RESTful Spring Controller
- Angular Frontends development
- Npm and NodeJs
- Java and JavaScript
- Reactive Programming with Typescript
- Json and yaml
- Jenkins Release Pipeline with Jenkinsfile
- Groovy Scripts
- Java 8 Functional programming
- JWT (JSON Web Token) for Microservices
- Maven Multi Module Projects
- DevOps Tasks
- Cloud Foundry
- Artifactory Repository
- Git and svn Repositories
- Svn to Git Migration

Introduction of Git-flow Workflow with GitLab  
In-memory data grid performance optimization  
XSS Cross Site Filter for Tomcat  
PKI Infrastructure  
Public Key, private Key communication  
One Time Password - OTP  
OpenID, Single Sign On - SSO  
Security Assertion Markup Language - SAML  
DirX and LDAP lookup  
Windows Active Directory Configuration  
Active Directory-Certification Service (AD CS)  
Apache Load balancing  
Tomcat Web Applications  
Shell Script programming  
Sonar for static code analyse  
OWASP Security Findings  
HTML5, CSS  
Spikes for Kotlin and Scala  
Kundenaccounts with Single Sign On  
Authentication / Authorisation  
Segregation Of Duties – SoD  
Agile Story Planung with JIRA  
Smartcard PKI Login  
Secure Coding  
Protection Requirements Assessment PRA  
User Management  
2-Factor Authentication  
Process definitions  
Coordination and Bug Tracking  
Agile Development with Scrum and Kanban  
Grooming, Planning, Retrospektive  
Remote communication over Circuit  
Project language Englisch  
Partly remote team

|                                             |                                     |
|---------------------------------------------|-------------------------------------|
| Realized with: Operatingsystems<br>Database | Windows, Linux<br>Oracle, H2, mySQL |
|---------------------------------------------|-------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developmenttools<br>Tools | Microsoft AD, AD CS, Visual Studio Code, Eclipse, STS, Java, Kotlin, Scala, Mockito, RxServer, Netty, Lombok, Cucumber, GitLab, YAML, Portecle, Ansible, JWT, Retire.JS, Git, svn, bash, Tomcat, Jetty, Entrust, Syncplicity, Splunk, Circuit, Subclipse, LDAP, DirX, X.509, WinSCP, puTTY, puTTYGen, Groovy, SQuirreL, eclEmma, cron jobs, Remedy, repIT, Lucidchart, Postman, ssh, ssl, Oracle, H2, Kanban, DevOps, Mailversand, Swagger, Netflix, Hystrix, Ribbon, Cobertura Code Coverage Report, cacerts, keytool, UML, ant, pageant, PMD, Checkstyle, Findbugs, Spotbugs, Sonar, Fiddler, SiMPL, Yo, npm, node, OWASP, jSoup, JIRA, conceptboard |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Financeportal for public financing - PIF

Role: Full Stack Developer

Branche: Bank, Fintech

References: Co. Dt. Pfandbriefbank/Capveriant in Munich/Unterschleißheim

Contact person Mr. Benedikt Girz, Mr. Sergiy Barlabanov

<http://www.pfandbriefbank.com>

<https://www.capveriant.com>

|      |         |           |      |          |
|------|---------|-----------|------|----------|
| from | 11/2017 | - 05/2018 | with | 7 months |
|------|---------|-----------|------|----------|

Java Microservices Architectur  
 Spring Boot, Spring Data  
 Java, JavaScript  
 Single Page Application  
 Angular Web-development  
 RESTful Interface  
 Frontend Tests with Protractor  
 JUnit Tests with Karma  
 Pacet Manager npm  
 Node.js non-blocking Runtime Engine  
 Distributed Source control with Git  
 Azure Cloud solutions  
 Automated ARM Templates  
 Azure Storage Account  
 Azure KeyVault Certificates  
 Powershell Azure Scripts  
 Setup of VSTS Build Pipeline  
 Automated Deployments with VSTS  
 VSTS for Tickets/Tacking  
 Multilanguage Application  
 Automatic E-Mail sending



SSL/HTTPS Certificates  
 YAML and JSON Dataformats  
 JSON Web Token JWT  
 Security Access Token  
 PaaS, SaaS as a Service  
 Kubernetes Container based Deployment  
 Docker for Container virtualization  
 MsSQL Docker Image creation  
 Transact-SQL (T-SQL) Scripts  
 Daten Migration with Flyway  
 Maven Build Management  
 DevOps tasks  
 Bash shell and Unix Scripts  
 Spike for Scala  
 Finanzportal for public financing  
 Complete new Development with a remote team  
 Agile Development with Scrum and Kanban  
 Grooming, Planning, Retrospective  
 Remote communication over TeamViewer, Skype, Slack  
 Project Speech Englisch  
 Offshore Team

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Windows, Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Database                        | Microsoft SQL Server, H2, HSQL, mySQL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Developmenttools                | Azure, Node.js, bash, npm, Eclipse, IntelliJ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Tools                           | IDEA, Visual Studio Code, Angular 5,<br>JavaScript, ES6, ESLint, Java8, Scala, Spring<br>Core, Spring Data, JPA, Hibernate, Mockito,<br>REST, Json, Maven, Notepad++, CSS, sublime,<br>SMTP, logback Logger, Putty, ssh, scp, Skype,<br>Slack, Teams, Ubuntu VM setup, CSRF,<br>Chocolatey Paketmanager, TCPView, MSSQL<br>Server, ARM Automatic Resource Management,<br>ARM Azure Resource Manager, DMA Data<br>Migration Assistant, SSMS SQL Server<br>Management Studio, SSMA SQL Server<br>Migration Assistant, Team Foundation Server<br>(TFS), Visual Studio Team Services (VSTS), Git,<br>Flyway, MsSQL, MySQL, Office365, zeplin.io,<br>Lucidchart, linoit, Typo3 Script Bitnami,<br>PrimeNG Rich UI Web<br>Components, Postman, cURL,<br>ModelMapper, querydsl, Bitnami<br>Matomo/Piwik, Apache FreeMarker, Spring<br>Quartz, GreenMail, ssh, ssl |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developmenttools<br>Tools | Azure, Node.js, bash, npm, Eclipse, IntelliJ<br>IDEA, Visual Studio Code, Angular 5,<br>JavaScript, ES6, ESLint, Java8, Scala, Spring<br>Core, Spring Data, JPA, Hibernate, Mockito,<br>REST, Json, Maven, Notepad++, CSS, sublime,<br>SMTP, logback Logger, Putty, ssh, scp, Skype,<br>Slack, Teams, Ubuntu VM setup, CSRF,<br>Chocolatey Paketmanager, TCPView, MSSQL<br>Server, ARM Automatic Resource Management,<br>ARM Azure Resource Manager, DMA Data<br>Migration Assistant, SSMS SQL Server<br>Management Studio, SSMA SQL Server<br>Migration Assistant, Team Foundation Server<br>(TFS), Visual Studio Team Services (VSTS), Git,<br>Flyway, MsSQL, MySQL, Office365, zeplin.io,<br>Lucidchart, linoit, Typo3 Script Bitnami,<br>PrimeNG Rich UI Web<br>Components, Postman, cURL,<br>ModelMapper, querydsl, Bitnami<br>Matomo/Piwik, Apache FreeMarker, Spring<br>Quartz, GreenMail, ssh, ssl |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Logistic for yard management - flow

Role: Full Stack Developer

Branche: Logistic

References: <http://www.43gis.com>

Co. 43gis in Munich.

Contact person Mr. Torsten Oltmanns, Mr. Clemens Weber

|                                                     |
|-----------------------------------------------------|
| from    03/2017    -    10/2017    with    8 months |
|-----------------------------------------------------|

- Microservices Architecture
- ReactJs frontend development
- Spring 4 and Java 8 backend development
- Single Page Applikation implementation
- Workflow implementation for Statemachine
- Redux Datastore connection
- REST communication
- Ajax and Promises Callback communication
- NodeJs and npm development
- Websocket Endpoint Messaging
- ESLint for code quality
- implementation of transactional Spring components
- Persistence with Hibernate
- Data migration
- SCM with Git and Gogs

Unit and integration tests  
 Jenkins Build control  
 JBoss, Wildfly Application Server  
 gradle and Maven build control  
 Certified Scrum Master PSM  
 Remote teams, Nearshoring  
 JIRA, confluence  
 Skype, TeamViewer  
 asterisk telefon conferences

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Windows, Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Database                        | Oracle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Developmenttools<br>Tools       | NodeJs, npm, ReactJS, JavaScript, Mocha,<br>ES6, JSX, Redux, PhantomJS, Jest, enzyme,<br>sinon, Lerna, karma, Lanterna, ESLint, Java 8,<br>Spring Core, Spring Data, JPA, Hibernate,<br>Mockito, JMX, Git Bash, Gogs (Go Git Service),<br>Jenkins, TeamViewer, REST, Json, Postman,<br>SQLDeveloper, ORACLE, MSSQLServer,<br>iBatis+Liquibase DB Migration, Eclipse Neon,<br>JBoss, Wildfly, Gradle, Maven, artifactory, Visual<br>Studio Code Editor, sublime, Atom, Notepad++,<br>JSP, JQuery, HTML, CSS, OpenLayers, jacoco,<br>Skype, Confluence, JIRA, Jing, Nagios, Bizagi<br>Modeler, Bamboo Paper, XMind Mindmaper,<br>realtimeboard, linoit, Scrum |

## Vehicle calculator for classic cars - ANDIE

Role: Full Stack Developer  
 Branche: Insurance  
 References: <http://www.aioinissaydowa.eu>  
 Co. AIOI in Munich/Ismaning.  
 Contact person Mr. Martin Brauner.

|      |         |           |               |
|------|---------|-----------|---------------|
| from | 10/2016 | - 03/2017 | with 6 months |
|------|---------|-----------|---------------|

Angular frontend development  
 Spring backend development  
 Development with Java 8 language features  
 REST call interface with Promises and Observables  
 Asynchron client/server communication with callbacks  
 Cross-Origin Resource Sharing (CORS) implementation with Spring  
 Protection against Cross Side Scripting (CSRF) with Spring  
 Grunt and Maven as Build Tool  
 Project modularization in Node.js, npm submodule  
 JSON and YAML file formats  
 Usage of HTML5 and CSS

Protractor E2E Tests  
 Jasmine Test Framework  
 Selenium Web Driver Tests with PhantomJS  
 Oracle VirtualBox for Vagrant virtualization  
 Migration with Postgre datenbase scripts  
 Complete new development in small team  
 Tariff calculation of car insurances for end customers  
 Customer registration with Double-Opt-in  
 Automated E-Mail delivery  
 Confluence Jira for communication  
 Process model with Scrum Team  
 Communication with Test Team

|                                 |                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Windows, Linux                                                                                                                                                                                                                                                                                                                                                  |
| Database                        | PostgreSQL                                                                                                                                                                                                                                                                                                                                                      |
| Developmenttools                | Flyway, Mockito, EasyMock, TSLint, Balsamiq,                                                                                                                                                                                                                                                                                                                    |
| Tools                           | Hibernate, Node.js, npm, Firebug, VS Code, CSS, HTML5, SonarLint, Spring Data, Spring Core, Spring Boot, GitHub, Angular2, Apache Commons, Docker, Vagrant, Jing, JIRA, Confluence, Json, YAML, SoapUI, Log4j, git, mvn, SourceTree Git Browser, Nexus, Jenkins, REST, JPA, DBeaver, PostgreSQL, Eclipse, STS, Java 8, putty, WinSCP, Notepad++, subethasmtplib |

## Telephone Number Porting with Java EE

Role: Backend Developer  
 Branche: Telecommunication  
 References: <http://www.consol.de>  
 Co. ConSol in Munich.  
 Contact person Mr. Thomas Michal, Mr Andreas Berger.

|      |                   |      |          |
|------|-------------------|------|----------|
| from | 04/2016 - 10/2016 | with | 7 months |
|------|-------------------|------|----------|

EJB backend development  
 Webservice REST Client/Server communication  
 Angular2 web development with TypeScript  
 TypeScript as abstraktion for JavaScript  
 Migration von EJB 2.1 nach EJB 3.0  
 Spring data for JPA repository access  
 TestNG components tests  
 Oracle WebLogic Application Server  
 Oracle PL/SQL Stored Procedures  
 RMI - Remote Method Invocation  
 Workflow engine

MSISDN migration over staging tables  
 JavaScript development  
 Node.js and npm for packages and modules management  
 JPA entity development with EclipseLink  
 Java 8 language features  
 Jenkins continues integration job administration  
 Citrus integrations tests  
 Maven multi project profile  
 Data migration of mass data  
 x.400 gateway and 2MRDB porting  
 CORBA Tuxedo connector  
 CORBA IDL  
 Scrum Master tasks  
 Telephone number porting  
 Import and export of telephone numbers for prepaid- and postpaid brands  
 Starter Pack SIM migration  
 Git branching strategies and pull requests  
 Atlassian tool suite for communication  
 Confluence Jira for processes  
 Bugtracking over ClearQuest  
 Load-tests  
 Realization within a scrum team  
 Coordination with domain experts  
 Communication with test-team

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Windows, Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Database                        | Oracle 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Developmenttools                | Eclipse, IntelliJ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Tools                           | Flyway, MyBatis Migration, Mockito, EasyMock,<br>Hibernate, maven-replacer-plugin, Node.js, npm,<br>Broccoli Plugin, TypeScript Eclipse Plugin,<br>Firebug, VS Code, SonarLint, TestNG, Spring<br>Data, Atlassian Suite, Bitbucket, Confluence,<br>JIRA, HipChat, ClearQuest, ClearQuest, GitHub,<br>AngularJS, Apache Commons, Docker, Jing,<br>JIRA, Confluence, Json, Log4j, Citrus, git, mvn,<br>ant, Nexus, Jenkins, Oracle WebLogic, SoapUI,<br>REST, JPA, SQLDeveloper, ORACLE, Eclipse,<br>Java 8, putty, WinSCP, Notepad++ |

## Multiple Assembly Planning & Process System - MAPP

Role: Backend Developer  
 Branche: Automotive

References: <http://www.gigatronik.de>

Co. Gigatronik in Munich.

Contact person Mr. Stefan Hanel, Mr. Michael Voitenleitner.

|      |                   |      |          |
|------|-------------------|------|----------|
| from | 09/2015 - 03/2016 | with | 7 months |
|------|-------------------|------|----------|

EJB development

Webservice REST Client/Server communication

JPA entity development with EclipseLink

JPA criteria API

Java 8 language features like Lambdas

Aspect oriented interceptors

DB modeling

Jenkins continuous integration job administration

JUnit Mockito tests and integration tests

Maven multi project profile

Usage of JRebel

JSON data exchange format

Data migration

JavaFx client development

JavaFx design with scene builder

Glassfish configuration

LDAP connection

SONAR metrics

FitNesse tests

Generic realization of MAPP for car production and planning of BMW vehicles

Part project coordination

Coordination with domain experts

Communication with Test-Team

Organization development meetings

'Clean Code' principles

Implementation and technical realization

Senior developer

Scrum Master tasks

Realized with: Operatingsystems    Windows, Linux

Database    Oracle 11

Developmenttools    SezPoz, Apache Commons, Jing, Jersey, Json,  
Tools    Log4j, EclipseLink, MQ, JRebel, mockito, SVN,  
mvn, FitNesse, Nexus, Jenkins, Sonar, Jubula,  
Glassfish, REST, JPA, SQLDeveloper,  
ORACLE, Eclipse, JavaFx, JAutodoc, LDAP,  
ScenicView, SceneBuilder, Java 8

**Online portal for ambulant, medical services - OPAL Web-Application**

Role: Backend-, Frontend-Developer

Branche: Health Insurance

References: <http://www.kvb.de>

Co. KVB in Munich. Contact person Mr. Matthias Geiss.

|      |         |   |         |   |      |           |
|------|---------|---|---------|---|------|-----------|
| from | 09/2012 | - | 07/2015 | _ | with | 35 months |
|------|---------|---|---------|---|------|-----------|

- Development of the online portal OPAL
- Extension of existing service interfaces
- Spring Core for Dependency Injection CDI
- Spring Web Flow for page navigation
- Spring Restful Webservice implementation
- Spring authentication and authorization
- Browser plausibility with JavaScript
- Buildmanagement with Gradle Buildsript in Groovy
- Continues Delivery with FeatureToggles
- Dynamic configuration of features with Framework 'Togglz'
- Webservice requests with SoapUI
- Agile DB migration with flywayDB
- Programming with Java 7
- Creation of SQL Scripts
- Serverside validation with Rhino Script Engine
- XSLT transformation of data to Sciphox and EHD format
- Further development of the JSP pages
- WebTests with JUnit and Selenium Browser Plugin
- Configuration of Jenkins Jobs for CI, NB and INT environment
- Timeline versioning of business change requests
- Usage of Jenkins plugins
- Configuration of Jenkins Jobs for CI, NB and INT environment
- Usage of Gradle and Maven as build tools
- Release management with maven release plugin
- Assembling and deployment with maven
- Maintainance of data in integration
- Development of ORACLE SQL statements for ABN and Production
- Defining of SQL scripts for DB handover process
- Maintainance of data with PL/SQL scripts
- File exchange formats .json and .yaml
- Asynchron processing over ONDES and KV-Connect interface
- Test Driven Development - TDD
- WebTests with JUnit and Selenium browser plugin
- Load Tests of Webservices with SoapUI
- Threading and Web Load-Tests with LoadUIWeb
- Remote access via Secure Shell (SSH) and odin
- Documentation of medical reports in the online portal (OPAL)
- Submission of electronic documentation to the KVB
- Export of data to Sciphox xml format

Support of the following medical documentations:

Allergologie, Balneophototherapie, Belastungs-EKG, Darmkrebsberatung, eDMP, DMP-Brustkrebs, Hautkrebsscreening, Koloskopie, Multiple Sklerose, Risikoprävention bei Kinderwunsch, Schmerztherapie, Sonografie Schwangerschaftsvorsorge, Tonsillotomie, Gastroskopie, Onkologie

New development of medical documentations like Hörgeräte, Molekulargenetik and Sozialpsychiatrie

Extension of the online documentation OPAL for medicians

Scrum with Daily Standup and Retrospective

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Windows 7, Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Database                        | Oracle 11g                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Developmenttools                | J2SE 7, STS - Spring Tool Suite, Kepler Eclipse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Tools                           | Java EE IDE for Web Developers, Tomcat 7, Jenkins Flow Plugins, Flyway, WinSCP, putty, Nexus, SONAR, ONDES, KV-Connect, Linux, PdfBox, Lombok, Gradle, Groovy, Spring MVC, SoapUI, Feature Toggly, Secure Shell (SSH), TLS, SSL, Burp Proxy, FoxyProxy, Hamcrest, Confluence, odin, DbSetup, flywayDB, maven release plugin, LoadUIWeb, Java 7, JavaMelody, xml-unit, JDedend, Spring Core, Spring Web Flow, Oracle SQL Developer, EJB 3.1, JPA, CriteriaAPI, SVN, Subclipse, Subversive, Tortoise, Firebug Firefox Addon, Selenium Plugin, Live HTTP headers Firefox Addon, EclEmma Eclipse Addon, CheckStyle, Maven, Tomcat, JUnit, XSLT, HttpUnit, DbUnit, Jenkins, joda-time.jar, jQuery, JavaScript Rhino Engine, JavaScript |

## **Web Wiki/Community Angeln.de (Fishing website)**

Role: Backend-, Frontend-Developer

Branche: Softwarehouse, Startup Company, Printmedia

References: <http://www.angeln.de>  
<http://www.woont.com>

Co. Woont in Munich. Contact person Mr. Dirk Wermuth.

|      |                   |      |           |
|------|-------------------|------|-----------|
| from | 10/2011 - 08/2012 | with | 11 months |
|------|-------------------|------|-----------|

Complete relaunch in a small team

Development of a Social-, Community-, Shop- Web-Application

Implementation of various Web-Applications around [www.angeln.de](http://www.angeln.de)

Web-Frontend, Shop, Editor and Admin realized as Web-Apps

Integration of social networking pages, linkage to other Web-Apps of the domain

Data import with SQL scripts and java importer



Development with Glassfish, Embedded Glassfish and Embedded Jetty  
 Creation of generic, reusable Ajax Wicket components  
 Extensive development of Ajax components for Web 2.0  
 Searching and sorting of datatables with Ajax  
 Usage of tables with paging navigator and ListView  
 Programming of image galleries with DataViews  
 Dynamic menus with changing contexts  
 Working with Wicket IModel variations like PropertyModel and CompoundPropertyModel  
 Layers Design Pattern - Design and call of interfaces to the backend  
 Implementation of SEO (Search Engine Optimization) techniques  
 Localized URL generation in various languages  
 Persistence layer with EJB, JPA and Criteria API  
 Geolocating of addresses with EJB Timer Bean  
 Google Maps integration with JavaScript Google API  
 Bean validation over various layers using JSR 303  
 Maintenance of the persistence service  
 Schema maintenance and enhancements for the database  
 Implementation of image and logo uploads  
 Cropping and converting of images with ImageMagick API  
 Image converting using im4java API for various web formats  
 Usage of Maven Multi Modules with personalized profiles  
 CSS adjustments and CSS generation using less compiler  
 Bug tracking system OTRS  
 Continuous integration and nightly build using Jenkins  
 Spring Dependency Injection Integration DI  
 CSS and JavaScript development and debugging with Firebug plugin for Firefox  
 Usage of EcEmma coverage eclipse plugin  
 Generic programming using java generics  
 Consequent usage of JUnit regressions tests, standard technologies and standard tools for development

|                                 |                          |
|---------------------------------|--------------------------|
| Realized with: Operatingsystems | Windows 7, Linux, Ubuntu |
| Database                        | PostgreSQL 9             |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developmenttools<br>Tools | J2SE 6, Eclipse 3.6.2 Helios, Glassfish Embedded, Glassfish 3.1.1, Embedded Jetty EJB 3.1, JPA, CriteriaAPI, Apache Wicket 1.5, Eclipse, SVN, Subclipse, Subversive, Firebug Firefox Addon, Live HTTP headers Firefox Addon, PrefBar Firefox Addon, AnyEdit Eclipse Addon, EclEmma Eclipse Addon, Maven, glassfish, Tomcat, Jetty, JUnit, Selenium, Jenkins, PMD, FindBugs, joda-time.jar, guava.jar, Skype, SQuireL SQL, PostgreSQL DB, gforge, tickspot, MindMap Freeplane, im4java, ImageMagick, GraphicsMagick, jQuery, less, JCrop, OTRS, Facebook, Twitter, YouTube, Google Maps API, fancyBox |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## **JEE Tariff Calculator and Webservice**

Role: Backend-, Frontend-Developer

Branche: Insurance

References: <http://www.lv1871.de>

Co. LV1871 in Munich. Contact person Mr. Alexander Gold.

|      |                   |      |           |
|------|-------------------|------|-----------|
| from | 07/2009 - 07/2011 | with | 25 months |
|------|-------------------|------|-----------|

Relaunch/Development of the existing Tariff Calculator software with Apache Wicket Framework

Development/Maintenance of the existing Tariff Calculator software with Apache Struts

Components with usage of Wicket HTML markup inheritance development

Development of advanced Gui functionality beyond the Wicket Extensions (like "close button" on tabulators)

Wicket Ajax component development

Development of company web based components

CSS and JavaScript development and diagnosis with Firebug Plugin for Firefox

Generating of path notation with 'bindgen' utility and @Bindable annotation for type save Wicket ID's

Extension handler for XStream and Castor marshallers for object to XML generators

Implementation of the BiPRO insurance webservice standard

Schema validation and WSDL generation

Design of complex element inheritance in XSD schemas

Usage of JAX-WS webservice framework

JAXB XML marshalling resp. XML serialization to java

Generic programming because of java generics usage

Loadtests with SoapUI against the webservice implementation

Design of the technical and logical validation of the webservice

Development of various XML namespaces

Implementation of various insurance tariffs

- Extensions on the existing tariff software
- Maintainance of the productive platform
- PDF generation with Apache FOP
- XSL transformation in several steps up till to printable XSL-FO documents
- Strict validation of XSL templates and XSL-FO elements
- Integration of the printable XSL-FO documents into the webapplication
- Refactoring of the exception handling
- Maintainance of the intranet and Wiki
- Consistent Usage of JUnit regression tests, standard technologies and standard tools for development
- Spring integration
- Usage of Core Spring for better Testing
- Usage of various JEE / J2EE Pattern like SessionFacade, ServiceLayer, Transfer Object Assembler, DTO, DAO, ...
- COBOL \*.cpy integration, calls to tariff calculator cores
- Automated XML printjobs
- Implementation of 'lucene' search engine functionality
- Usage of EclEmma Coverage Eclipse Plugin

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Windows 7, Windows Vista, Windows XP, Linux, AIX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Database                        | HDB, DB2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Developmenttools<br>Tools       | J2SE 6.0, Eclipse 3.x, Jetty6, Tomcat6, Live HTTP headers, Firebug, Citrix, STS, Subversion SVN, CVS, WinSCP, PuTTY, Extreme Planner, Firefox, Opera, Safari, Windows 7, Windows Vista, Windows XP, commons beanutils, xalan Parser, xerces Parser, JUnit, httpUnit, Spring, Bindgen, ant, maven, Jenkins/Hudson Nightly Build Tool, Emma Test Coverage Frameworks, JAX-WS Webservice Framework, SoapUI, UML Enterprise Architect, XMLSyp, Notepad++, Bouncy Castle Cryptography Library, JDom, xstream serialize XML, Apache Struts, Apache Wicket, Apache log4j, Apache FOP, BiPRO, JAXB, WSDSL2Java, cygwin |

## Smartphone Web and App Development

Role: Backend-, Frontend-Developer

Branche: Entertainment

References: [http://www.androidzoom.com/android\\_applications/trivial+dice/by\\_matching](http://www.androidzoom.com/android_applications/trivial+dice/by_matching)  
<http://www.m.hopf-it.de>

|      |                   |      |         |
|------|-------------------|------|---------|
| from | 09/2010 - 10/2010 | with | 1 month |
|------|-------------------|------|---------|

Web app Development for Smartphones

- Usage of iui - User Interface (UI) Framework from Google
- Development of the own homepage in mobile format
- Development of a Simple dice which rolls in 2D graphic, if you tap the screen
- Game development: Trivial Dice
- Conception of the layout
- Design of the graphic
- Development of the statemachine
- External event handling
- Consideration of different resolutions
- Testing of various devices with emulator
- JUnitTest suite for the App
- Lite version with reduced feature set
- Pro version with different access rights
- Playing of audio file
- Familiarization with the Anroid SDK and Google iui, JQTouch Framework
- Multi lingual description
- Versioning in Android Market

|                                 |                                                       |
|---------------------------------|-------------------------------------------------------|
| Realized with: Operatingsystems | Windows 7                                             |
| Database                        | -                                                     |
| Developmenttools                | J2SE 6.0, Eclipse 3.x, Google iui Framework,          |
| Tools                           | JQTouch Framework, Android SDK, Eclipse Plugin, JUnit |

## **JEE Webapplications for Ring Back Tones**

Role: Backend-Developer

Branche: Telecom

References: <http://www.ect-telecoms.de>

Co. ect-telecoms in Munich. Contact person Mr. Robert Parkes, Mr. Hans Huber, Mr. Ahmed Ferdous.

|      |                   |      |           |
|------|-------------------|------|-----------|
| from | 09/2008 - 06/2009 | with | 10 months |
|------|-------------------|------|-----------|

- Creation of web applications with Tomcat, JSP, Spring and Hibernate
- HTTP Authentication with Basic and Spring XwsSecurity method
- Porting respectively migrating the integration layers from Oracle to HSQL database
- Porting of Oracle schemas and data to HSQL database
- Realization of a standalone reference web client as a presentation application
- Implementation of a Soap Services with JAXB to upload binary content
- Configuration of ant scripts and make them runnable under LINUX and Windows
- Extension of the Soap core component with new features
- Publishing the new features as web service
- Porting the Codehouse XFire web service architecture to Spring WS respectively Apache CXF Framework
- Mobile Communication over USSD Service Codes and XML-RPC.
- MDA UML transformation to XSD

XSD schema definition for automatic generation of the model and soap classes  
 JAXB XML marshalling respectively XML Serialization to Java  
 Basic Apache webserver configuration  
 Remote installation and deployment onto customer servers  
 Creation of Spring, maven and ant configurations  
 Error tracking with Jira as bug tracking system  
 Initialization and introduction of Apache Continuum as Nightly Build process, JUnit  
 Tests, Nightly Build, SVN Tags and Branches, measure of test coverage, Intranet  
 Wiki.  
 Load tests for Soap web services with soapUI  
 junit regression tests  
 Consequent usage of JUnit regression tests, standard technologies and standard  
 tools in the development  
 Team communication and documentation completely in English

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Windows; Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Database                        | Oracle, HSQL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Developmenttools                | J2SE 5.0, GIMP PictureEditor, ColorZilla                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Tools                           | Colorpicker, FireFTP, Notepad++,<br>Cygwin, HSQL, ORACLE10g, Eclipse 3.4,<br>Tomcat5.5, Live HTTP headers, SquirrelL<br>DBViewer,<br>SQLDeveloper, SQLPlus, DbVisualizer, VNC, 7-<br>Zip, Tortoise, Subversion SVN, WinSCP,<br>PuTTY, Jira Bugtracking,<br>Clearspace, Thunderbird, Firefox, Ace<br>WINScreen, Windows XP, Linux Ubuntu, Spring,<br>Hibernate, ehcache, OSGi, commons,<br>myfaces, xalan, xerces, JUnit, httpUnit, ant,<br>maven, Codehouse XFire, Apache CXF,<br>XML-RPC, redstone XML-RPC library, GSM<br>Services, EMF (Eclipse Modelling Framework),<br>GMF (Graphical Modelling Framework), SOAP,<br>Spring WebService, jstl, acegi security,<br>JavaScript, JavaScript Debugger Venkman,<br>clover and emma test coverage Frameworks,<br>Apache CXF Webservice framework, soapUI,<br>UML Enterprise Architect, StarUML |

## **J2EE accounting system for health insurance (nPROD)**

Role: Developer

Branche: Health Insurance

References: <http://www.azh.de>

Co. VSA/azh in Munich. Contact person Mr. Weber, Mr. Herrmann.

|                                       |
|---------------------------------------|
| from 04/2006 - 08/2008 with 28 months |
|---------------------------------------|

New product development of an automatic accounting system for health- and auxiliary material

Analysis and design with UML 2.1

Requirement analysis and review with business and domain experts

Use Case description with the CASE Tool MagicDraw

MDA usage for the persistence layer

Usage of a persistence framework (Marvin) for the modelling of the DB

Modelling of GUI workflows with the CASE Tool MagicDraw

Usage of a GUI workflow framework

SWING masks with GidBagLayout

Extensive usage of CVS (Tagging, Branching, Merging)

Rule based approach for frequently changing requirements

Migration of the project to the Spring application framework

Migration of the existing data to the new structure using CSV files

Adaptation of the data structures for the data import

Data import from ABDA pharmacy position data

Electronic data exchange (DTA) with health insurance company

JUnit-Tests for the business logic and the GUI

Quality ensured with 80% JUnit test coverage and CheckStyle coding conventions

Maintenance and rollout

Attendance of the project from beginning to the end (product version 1.0)

Incremental approach on database changes with update scripts

Practical experience with a reference database

Automatic price calculation for diverse health insurance funds in southern Germany

Tasks: master data, price calculation, prescription position management, master agreement, DTA key management.

Development of an account software nPROD for automatic recipe processing.

Communication with an understaffed domain department.

Quality enhancement with coupling- and dependency metrics

|                                          |
|------------------------------------------|
| Realized with: Operating systems Windows |
|------------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developmenttools<br>Tools | J2SE 5.0, Eclipse 3.3, Jbossrules, Drools Regel-Engine, RuleEngine, Castor as XML Mapper, Clover for Testcoverage, JUnit for Programtests, CheckStyle, ANTLR (ANother Tool for Language Recognition) Parser Generator, Ant, JReport Presentationtool/FOP, Marvin proprietary Persistence Framework and Spring Persistence Framework 2.0, GUI Workflow Framework, IntelliJ 4.5, Jboss 4, MagicDraw 14, DB2-Client 7.2.0, Citrix Emulator, DBVisualiser, Toad, Squirrel, SQL Workbench, WinCVS 2.0, TortoiseCVS, Firefox, Thunderbird, Nvu (Html-Editor), cygwin Unix Emulator, PuTTY, WinSCP |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## J2EE WebShop

Role: Developer

Branche: eCommerce

References: <http://www.publicWEBshop.com>

|      |                   |      |          |
|------|-------------------|------|----------|
| from | 12/2005 - 02/2006 | with | 3 months |
|------|-------------------|------|----------|

Webapplication to run an online web shop.  
 Generic webshop for all kinds of businesses.  
 Freely configurable with many settings and preferences.  
 JSF Technologie, Apache MyFaces/Tomahawk .  
 Axis WebService JAX-RPC interface for B2B SOAP communication.  
 EJB, JSP and Servlets.  
 Struts and tiles.  
 Apache Commons.  
 AJAX asynch. communication on the client tier.  
 Object persistance with Hibernate.  
 Struts and tiles.  
 Logging with Log4j.  
 JavaScript and cascading stylesheets (CSS).  
 Planning and maintaining of a MySql database.  
 UML analysis and design.  
 XML for configuration files.  
 Multilingual for international purpose.  
 HTML, XHTML and Flash as presentation technique.  
 Tests with Jakarta JMeter and JUnit.  
 Future: XSLT and XSL-FO for PDF generation.  
 Future: EJB/JBoss/Spring.

|                                 |                |
|---------------------------------|----------------|
| Realized with: Operatingsystems | Windows, Linux |
|---------------------------------|----------------|

|                           |                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developmenttools<br>Tools | Eclipse 3.1, Enterprise Architect, XMLSpy<br>Enterprise Edition, JBoss, Tomcat 5.5, Struts,<br>Ant, JUnit, Macromedia Flash and Dreamweaver,<br>div. Eclipse Plugins, MySQL 5 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Trainings and Consulting

Role: Developer

Branche: All areas

References: here are the biggest companies:

Co. SUN Micosystems in Munich. Contact person Mr. Hettmannsberger.

Co. Accenture in Munich. Contact person Mrs. Steffen.

Co. Hilf in Oberhaching/ Munich. Contact person Mr. Hilf.

Co. AS Systeme in Stuttgart. Contact person Mr. Buehler.

Co. Tria in Munich.

|      |                   |      |           |
|------|-------------------|------|-----------|
| from | 04/2004 - 11/2005 | with | 19 months |
|------|-------------------|------|-----------|

J2EE

EJB, JSP and Servlets.

Java , SWING.

Webapplications.

EAI integration with differant interfaces.

Message queue connetions over JMS api.

Webservices and SOAP.

JAX-RPC.

UML analysis and design.

XML and XSLT.

J2EE and GOF design pattern.

UML for embedded systems.

OOAD object oriented techinques.

XML and XSLT.

Xalan and Xerces parser event processing.

Developing of course scripts.

Englisch courses.

|                                 |                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Windows, UNIX Solaris                                                                                                                                                                                                                                                                                                                          |
| Developmenttools<br>Tools       | Eclipse 3.0, NetBeans, JBuilder, Rational XDE<br>and Rose 7.5, Together Controll Center,<br>Poseidon, Enterprise Architect,, XMLSpy<br>Enterprise Edition 2006, SUN ONE, BEA,<br>JBoss, Tomcat, Struts, Ant, JUnit, XDoclet, div.<br>Eclipse Plugins, JAD Decompiler, Obfuscation,<br>Xalan and Xerces Parser, Pointbase, MySQL,<br>Cloudscape |

## Java-COM Bridge over JNI



Role: Developer

Branche: Insurance

References: Kassenärztliche Vereinigung Bayern (KVB) in Munich.

Contact person Mr Dietl.

|      |         |   |         |      |          |
|------|---------|---|---------|------|----------|
| from | 02/2004 | - | 03/2004 | with | 2 months |
|------|---------|---|---------|------|----------|

Refactoring and Reengineering of an existing JNI Interface.

Java Native Interface (JNI)..

Realisation with java and C++.

Analysis and design of the existing interfaces.

Changing the interfaces from synchron to asynchron.

Design of the asynchronous communication model .

UML analysis and design.

Use of GOF design pattern .

J2EE integration of JBoss with MS Office.

JNI API for distributed objects, realised in C++.

ANSI C and C++.

Documentation.

Consulting and project leadership.

Coaching of the employees.

|                                 |                                               |
|---------------------------------|-----------------------------------------------|
| Realized with: Operatingsystems | Windows XP                                    |
| Developmenttools                | JNI, COM, Visual C++ 6.0, Eclipse 3.0, JBoss, |
| Tools                           | Centro and JACOB JNI Tool                     |

## Consulting, Trainings and Courses

Role: Developer

Branche: All areas

References: here are the biggest companies:

Co. SUN Microsystems in Munich. Contact person Mr. Hettmannsberger.

Co. Soluzione in Munich. Contact person Mr. Roda.

Co. Hilf in Oberhaching/ Munich. Contact person Mr. Hilf.

Co. New Horizon in Hamburg. Contact person Mrs. Schmitt.

Co. Bosch in Lohr and Hildesheim.

Co. Amatech in Germering.

|      |         |   |         |      |           |
|------|---------|---|---------|------|-----------|
| from | 10/2001 | - | 01/2004 | with | 27 months |
|------|---------|---|---------|------|-----------|

J2EE

Workshops for the whole Java certification program.

Java Workshop.

Webapplications and E-Commerce architectures.

EJB, JSP and Servlets.

C#.

UML analysis and design.

XML and XSLT.

Microsoft MCSD courses.  
 ASP and .NET.  
 ANSI C and C++.  
 Using modern presentationtools.  
 Developing of course scripts.  
 Englisch courses.

|                                 |                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Windows, UNIX                                                                                             |
| Developmenttools                | Visual Studio 6.0, Eclipse, NetBeans, JBuilder,                                                           |
| Tools                           | Rational XDE and Rose 7.5, Together Controll<br>Center6, .NET, SQL Server, XMLSpy, SUN ONE,<br>BEA, JBOSS |

## Internet application for an billingsystem

Role: Developer

Branche: Bank, Billingsystem

Reference: Landeszentralbank (LZB) Bayern in Munich. Contact person Mr.  
 Stuckmann.

|                                       |
|---------------------------------------|
| from 05/2000 - 09/2001 with 17 months |
|---------------------------------------|

Analysis and design with UML and Rational Rose.  
 Refactoring, Reengeneeing of UML Modells, realised with Java and C++.  
 Generic modelling with RUP. Generation of artifacts.  
 JavaBeans and serverside Enterprise JavaBeans development.  
 XML documents with Schema spezifikation.  
 JSP pages and Java Servlets development.  
 Java Script programing for all browsers.  
 Webapplications and Webcomponents.  
 JSP Engine configuration and support.  
 C++ Framework development with STL.  
 JNI API for destributet objects in C++.  
 Crypt with Legacy Systems from RSA and PKI.  
 Webserver IPlanet and IIS configuration and administration.  
 Complex SQL-Queries.  
 SWIFT SIPN and InterActBrowse connection.  
 SWIFT Webstation support and configuration.  
 Performant PL/SQL-prozedures.  
 Administration of ORACLE database.  
 Team-Software development with versioning-software Merlin.

|                                 |                                 |
|---------------------------------|---------------------------------|
| Realized with: Operatingsystems | Windows NT 4.0, IIS WebServer   |
| Developmenttools                | Visual C++ 6.0, JBuilder 3.5    |
| JSP Engines                     | Apache TomCat, Allaire JRun 3.1 |
| Databases                       | ORACLE 8.x, DB2                 |

|                   |                                                                   |
|-------------------|-------------------------------------------------------------------|
| Emulations, Tools | Rational Rose 7.5, Telnet, FTP, SQL-ODBC-Viewer, SWIFT Webstation |
| Interfaces        | JNI, XML, SWIFT-SIPN                                              |

## Administration and support of a Stock-Trading and Portfolio-System

Role: Developer

Branche: Bank, Stock

References: Co. FMIS - Hypovereinsbank in Munich. Contact person Mrs Sterr.

|      |       |   |         |      |           |
|------|-------|---|---------|------|-----------|
| from | 02/99 | - | 03/2000 | with | 13 months |
|------|-------|---|---------|------|-----------|

Administration of stock transactions.

Developing of ORACLE databaseprocedures.

Coordination of projects.

Stockorder validation: orderentrance, routing to the stock markets and protocolling in ORACLE databases.

Developing of automatic flows to store new customers.

Implementing and coordination of the „Ertragnisaufstellung/Jahressteuerbescheid“.

Objectoriented interfaceprograming to BossCube and Xetra

Interfaceprograming between UNIX and AS400.

Objectoriented progaming of Stockedit GUI's.

Complex SQL-Queries.

Performante PL/SQL-prozedures.

Support of Windows programs for Broker-Controlling.

Support of Customerdata.

Administration of emmisionen.

Daily Stockbuisness.

Year 2000 exchange.

32 Bit exchange.

|                                 |                                                    |
|---------------------------------|----------------------------------------------------|
| Realized with: Operatingsystems | Windows NT 4.0, UNIX Sun Solaris                   |
| Developmenttools                | Visual C++ 5.0, Centura SQLWindows 1.5             |
| Databases                       | ORACLE 8.x, 7.x, SQLBase                           |
| Emulations, Tools               | Hostlink, Unixlink, Telnet, FTP, SQLTalk, SQLPlus. |
| Interfaces                      | BossCube, Xetra.                                   |

## Developing an Internet-Applet for Credit Scoring

Role: Developer

Branche: Finance and Insurance

References: Co. NSE in Munich. Contact person Mr. F. Nerb.

|      |       |   |       |      |           |
|------|-------|---|-------|------|-----------|
| from | 05/98 | - | 02/99 | With | 10 months |
|------|-------|---|-------|------|-----------|

Internet Applet and Server for a Bausparkasse.

Client/Server application for usage within WWW.

Customerown objectoriented classlibrary (ROI -> CORBA-subamount).

Middleware and IDL programing

Tests with all Internet-browsers, versions and plattform.

Bauspar-calculatingcore and surfaces

Teamprograming (Sourceadmin).

Databaseadmin, access to ISAM databases.

Develpoing of objectoriented GUI user interfaces.

Printing over DDE interface (Reportdesigner).

Documentation.

|                                 |                                                                    |
|---------------------------------|--------------------------------------------------------------------|
| Realized with: Operationsystems | Windows NT 4.0                                                     |
| Development tools               | Java 1.1 and Visual C++ 5.0                                        |
| Databases                       | ISAM Database                                                      |
| Tools, Libraries                | ROI, Middelware, Customerown objectoriented Classlibrary, Internet |

## Developing a database-admin-program

Role: Developer

Branche: Softwarehouse

|      |       |   |       |      |          |
|------|-------|---|-------|------|----------|
| from | 01/98 | - | 04/98 | with | 4 months |
|------|-------|---|-------|------|----------|

Spezifikation of the Softwarefeatures.

Support and admintools for Access database.

Developing of GUI user Interfaces.

Database administration over ODBC and RDO.

MS DAO interface with JetEngine.

SQL query language.

Documentation.

|                                 |                                                   |
|---------------------------------|---------------------------------------------------|
| Realized with: Operatingsystems | Windows NT 4.0                                    |
| Developmenttools                | Visual Basic 5.0                                  |
| Databases                       | Access 7.0                                        |
| Tools, Libraries                | MS-JetEngine, DAO, RDO, ODBC, InstallShield, MSDN |

## Installation of Network

Role: Developer

Branche: Advertising

References: Co. MPS in Munich. Contact person Mr. Hunsinger.

|      |       |   |       |      |          |
|------|-------|---|-------|------|----------|
| from | 11/97 | - | 12/97 | with | 2 months |
|------|-------|---|-------|------|----------|

Planning and installation of a Windows Network (5 PC's).

Connection protocol NetBUI and Ethernet 10 MBit.

Using a Networkhub.

Installation of Internetaccounts.

|                                 |                                    |
|---------------------------------|------------------------------------|
| Realized with: Operationsystems | Windows 95, Windows NT Server 4.0. |
| Technique                       | RJ 45 twisted pair wires.          |

## Programing of customer spezific softwaresolutions

Branche: Lasertechnique

References: Co. Baasel in Starnberg. Contact person Mr. Wutz.

|      |       |   |       |      |           |
|------|-------|---|-------|------|-----------|
| from | 03/96 | - | 11/97 | with | 21 months |
|------|-------|---|-------|------|-----------|

Spezifikationen of the Softwarefeatures.

Customervisits and discussions.

Developing of GUI Interfaces.

Reading of techniquial values from datbases.

Serial communication.

SPS connection of mashines in processtreets.

Connecting of several production pc's with NT 4.0 Server.

Connection to Windows NT pc's to an UNIX-mainframe over FTP protocol.

Administration of Windows NT.

Documentations.

Installations.

Service travels.

|                                 |                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------|
| Realized with: Operatingsystems | Windows NT 4.0, Windows 95, Windows 3.x                                                                  |
| Developmenttools                | Visual Basic 5.0, C, Visual C++                                                                          |
| Databases                       | Access 7.0, FoxPro                                                                                       |
| Tools, Libraries                | MS-JetEngine, DAO, RDO, ODBC, Custom Controls ActiveX from Crescent, OnNet FTP, InstallShield, MFC, MSDN |

## Developing of a Windows application to measure the pollution of Cars

Role: Developer

Branche: Automobil

References: Co. Bosch in Plochingen. Contact person Mr. Roth.

|      |       |   |       |      |          |
|------|-------|---|-------|------|----------|
| from | 07.94 | - | 12.94 | with | 6 months |
|------|-------|---|-------|------|----------|

Working in a Softwareteam (5 people).

Using MS SourceSave.

Extensiv use of Windows API.

Design and development of GUI dialogs.

Serial communication to the measure devise.

Print of Data on the printer.

Protocolling into the database.

Test and support.

|                                 |               |
|---------------------------------|---------------|
| Realized with: Operatingsystems | Windows 3.11  |
| Developmenttools                | Borland C/C++ |

|                  |                                        |
|------------------|----------------------------------------|
| Databases        | Access 2.0                             |
| Tools, Libraries | SourceSave, Installshield, Windows API |

## Developing an Satellite Network Management System

Role: Developer

Branche: Telecommunication

References: Co. ANT Nachrichtentechnik in Backnang. Contact person Mr. Fröschle.

|      |       |   |       |      |          |
|------|-------|---|-------|------|----------|
| from | 02.94 | - | 07.94 | with | 6 months |
|------|-------|---|-------|------|----------|

Analysis of the problem in cooperation with the users.

Documentation of the features.

Design and development of GUI-Modules.

Using an objectoriented database

Developing a Client/Server application.

Implementation and documentation in english.

|                                 |               |
|---------------------------------|---------------|
| Realized with: Operatingsystems | UNIX Solaris  |
| Developmenttools                | vi Editor     |
| Databases                       | Spectrum      |
| Tools, Libraries                | C++, OS Motif |

## Application to administrate files in a Novell-Network.

Role: Developer

Branche: Cartographie

|      |       |   |       |      |          |
|------|-------|---|-------|------|----------|
| from | 08.92 | - | 01.93 | with | 6 months |
|------|-------|---|-------|------|----------|

Developing of menue user interfaces in "C".

Interface-convertierung in sveral formats (HPGL).

Developing a Network-Multiuser-Tools.

|                                 |             |
|---------------------------------|-------------|
| Realized with: Operatingsystems | Windows 3.1 |
| Developmenttools                | Microsoft C |